

TORCH

Revision

TORCH

OVERVIEW

- Stands for **T**oxoplasmosis, **O**ther pathogens, **R**ubella, **C**MV and **H**erpes simplex viruses
- Other pathogens include: **S**yphilis, **H**IV, **H**BV
- **Trans-placental transmission (Vertical)**
- Out come ranges from healthy to fetal death
- Similar clinical manifestations



TORCH INFECTION

Early outcomes of TORCH infection

- Abortion, Congenital malformation
- Still born, Neonatal clinical illness
- Healthy or asymptomatic infection

Late clinical outcomes

- Deafness due to neuritis
- Visual loss due to chorioretinitis
- Mental retardation
- Cerebral palsy.
- Seizure disorder
- Obstructive hydrocephalus



TORCH INFECTIONS

Clinical manifestation

- Respiratory distress
- Petechiae, purpura due to endothelial injury
- Hepatomegaly, Splenomegaly, Jaundice
- Seizure disorder, Hydrocephalus
- IUGR, Small for gestational age
- Congenital malformations



TORCH INFECTION

Special features

- Cataract and PDA in **RUBELLA**
- Cicatrization (heal by scar formation) in **Hernia**
Zoster
- Snuffles, rash and Hutchinson's triad in **sypphilis**
- Brain calcifications in **toxoplasmosis** and **CMV**
- Chorioretinitis in **toxoplasmosis**
- Hydrocephalus in **CMV** or **Toxoplasmosis**



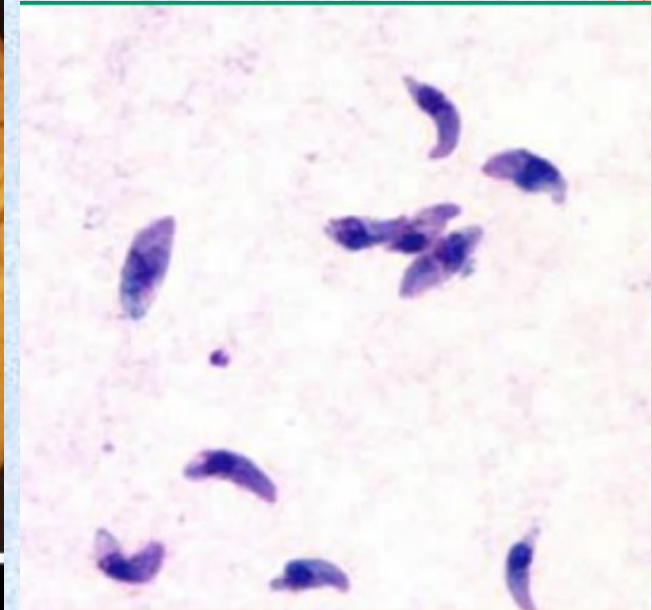
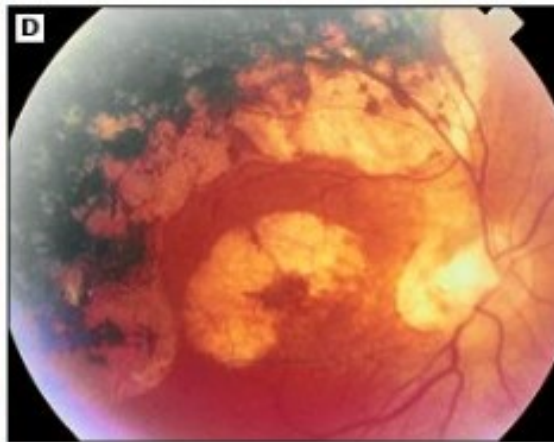
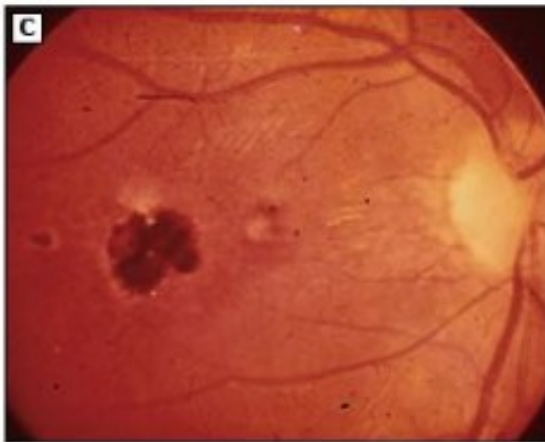
TORCH INFECTIONS

Toxoplasmosis

- Maternal infection due to contact to **feces of cats and eating raw vegetables and meat**
- Early gestational infections lead to **abortions and or congenital malformations**
- **Hydrocephalus, chorioretinitis, Hepatomegaly, splenomegaly and seizure are common clinical manifestations.**
- Chorioretinitis is the leading cause of blindness if not treated early



CHORIO-RETINITIS IN CONGENITAL TOXOPLASMOSIS

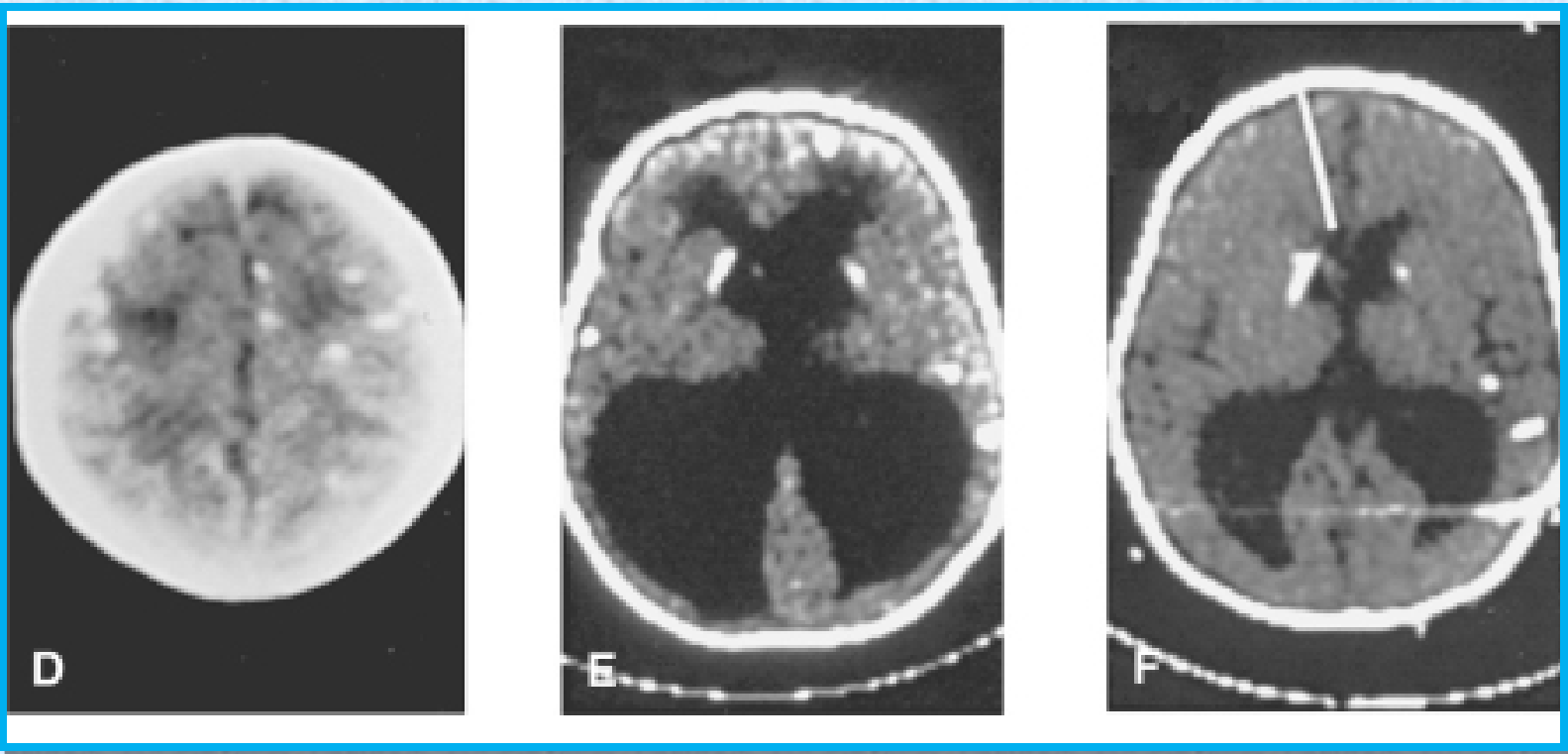


TOXO--TACHYZOITES

- (A) Severe, active retinochoroiditis.
- (B) Peripheral retinochoroiditis.
- (C) Central, healed retinochoroiditis.
- (D) Multiple chorioretinal scars.

Panels A, B, and C reproduced from: Centers for Disease Control and Prevention.

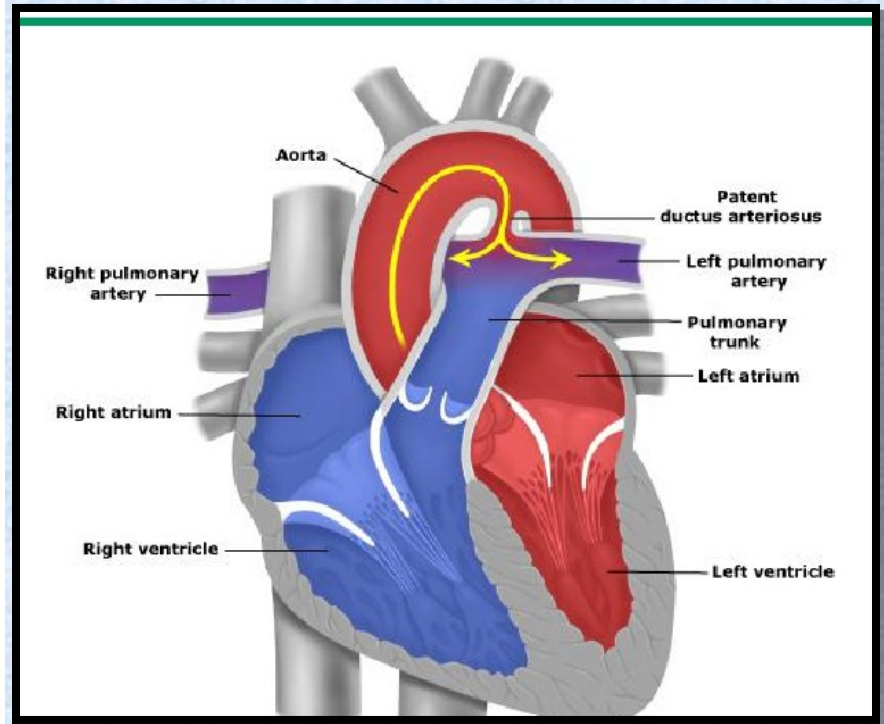
TORCH INFECTIONS



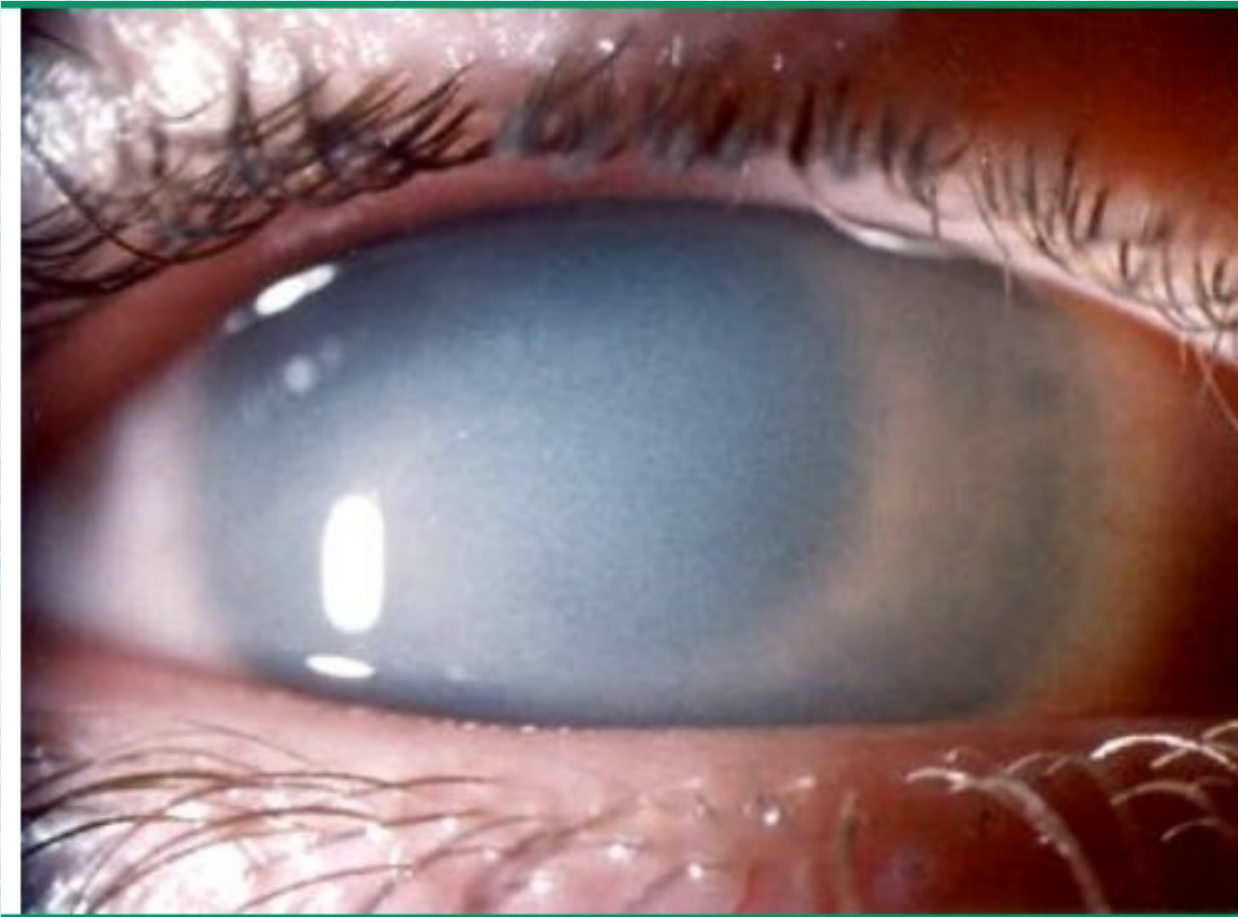
CT scan of a neonate with congenital toxoplasmosis. Note multiple brain calcifications and hydrocephalus, brain atrophy and VP shunt tube in the last CT slide.



CONGENITAL CATARACT AND PDA IN CONGENITAL RUBELLA



CORNEAL CLOUDING



CONGENITAL RUBELLA SYNDROME



Blueberry muffin lesions due to TORCH infection.
It is due to skin hematopoiesis.



VARICELA ZOSTER INFECTION IN UTERO



Notice the **ciatrical** scaring of the abdomen with bilateral deformity of both limbs.



TORCH INFECTIONS

Syphilis congenital infection

- Infection increase at **late gestation**.
- Most are asymptomatic at birth
- **Early** signs include skin rash, snuffles (discharging through nostril), HSM and bones changes.
- **Late** signs are Hutchinson's triads, saddle nose, Olympian (beyond what is usual) brow, saber (sword like) shin etc
- **Hutchinson's triad: keratitis, deafness, peg teeth**



SYPHILIS OSTEITIS AND SNUFFLES



Congenital syphilis: Snuffles

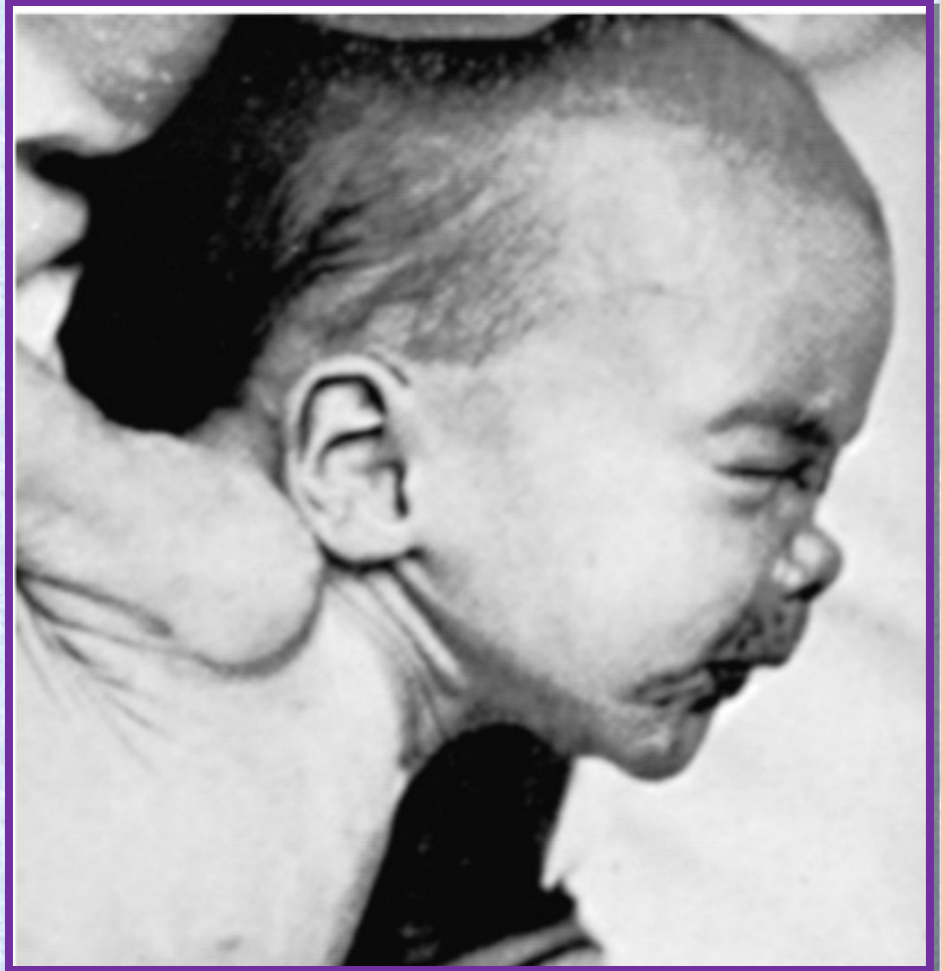


TORCH INFECTIONS

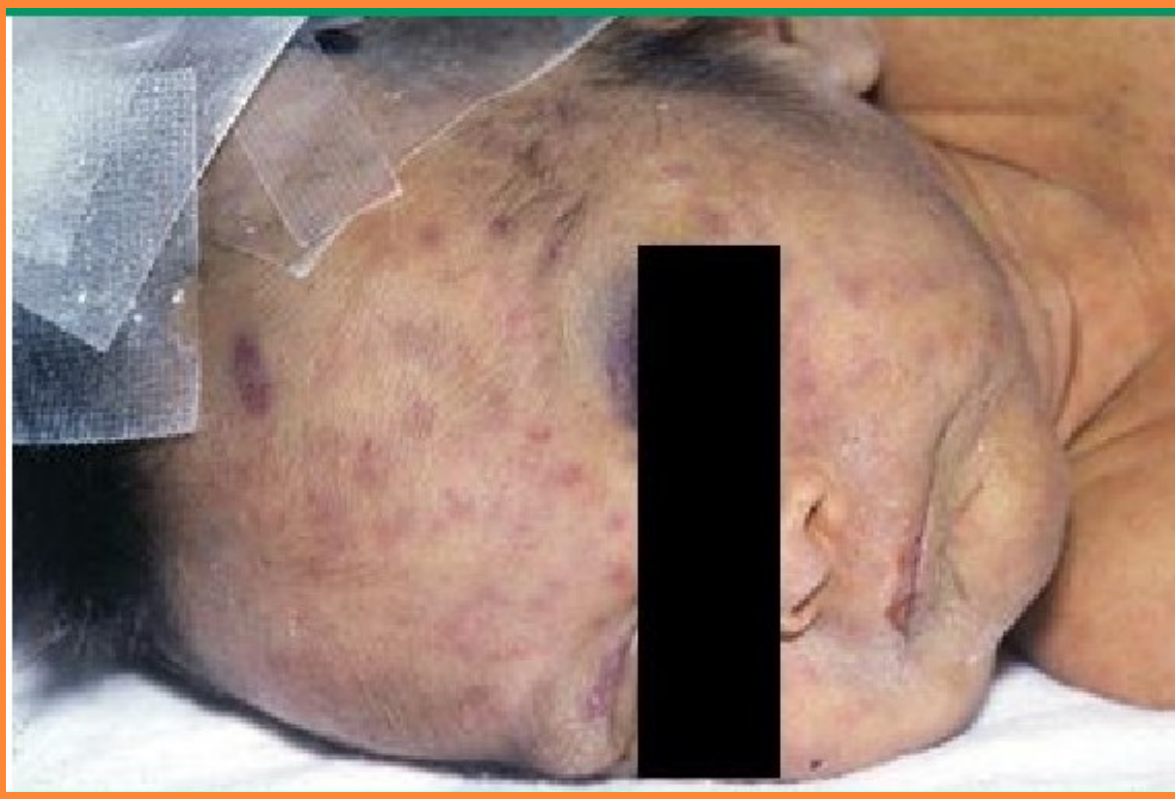


Papulosquamous skin lesion in congenital syphilis.

PEG TEETH AND SADDLE NOSE A COMPLICATION OF SYPHILIS



CONGENITAL CMV INFECTION WITH PETECHIAE AND PURPURA. CT-SCAN SHOWING CALCIFICATIONS



TORCH INFECTIONS

Diagnosis

- Complete blood count (Anemia, thrombocytopenia, leukopenia or leukocytosis)
- Serology test- VDRL for syphilis
- Dark field microscopy for syphilis
- IgG and IgM for Toxo, CMV, herpes and rubella
- CSF analysis (protein, glucose, WBC)
- Viral culture from CSF, urine and or blood
- CT scan or Brain Ultrasound



TORCH INFECTION

Treatment

- Syphilis- Pencillin sensitive
- Mother should be treated including the partner
- Toxoplasmosis- spiramycin for the mother
- Pyrimethamine/sulfadoxine for the baby
- CMV, HSV, HZV- Acyclovir or Ganciclovir
- Management of complications (hydrocephalus, congenital heart disease, cataract etc)



TORCH INFECTION

Prevention

- Maternal immunization (**MMRV, Hep**)
- Pregnant ladies should avoid contact to cats, raw meat and vegetables to decrease toxoplasmosis
- Anti-retroviral for pregnant ladies and neonates to decrease **HIV** transmission
- Safe sex to avoid **HSV, HBV, HIV** and ***syphilis***
- **HBV** vaccine and immunoglobulin for the baby to decrease HBV transmission





OSTEITIS OF CONGENITAL SYPHILIS



HUTCHINSON'S TEETH OF CONGENITAL SYPHILIS

